

6. Applications of Integrals

6.4. Work

6.4 a Learning Objectives:

- I can understand the definitions of work and force.
- I understand how to use integration to describe work.
- I can describe the work done in pulley systems and springs.

Definition 6.4.1

Force is any influence that causes an object to undergo a change in speed, direction, or shape. Force has magnitude and direction. We say that an object with mass, m , and acceleration, a , has force

$$F = ma$$

A body moves a distance, d , along a straight line after being acted by a force of constant magnitude, F , in the direction of motion, we define the work, W , done by the force on the body with the formula

$$W = Fd$$

6.4.1. Units

In this class you will encounter both metric system and imperial system.

In metric,

- Distance should be converted to meters, m, to work with other units.
- Mass should be converted to kilograms, kg, to work with other units.
- Time is measured in seconds, s.
- Force is measured in Newtons, N, where 1 N is equivalent to 1 kg m s^{-2} .
- Work is measured in Joules, J, where 1 J is equivalent to 1 N m.
- Acceleration due to gravity is $g = 9.8 \text{ m s}^{-2}$.

In imperial system,

- Distance is measured in feet, ft
- Mass is not normally given. Instead weight (which is a force) is given.
- Force is measured in pounds, lb.
- Work is measured in foot-pounds, ft-lb.

Example 6.4.1

Determine the amount of work done.

- a. A 5 kg book is lifted 4 m.
- b. A 5 lb book is lifted 4 ft.

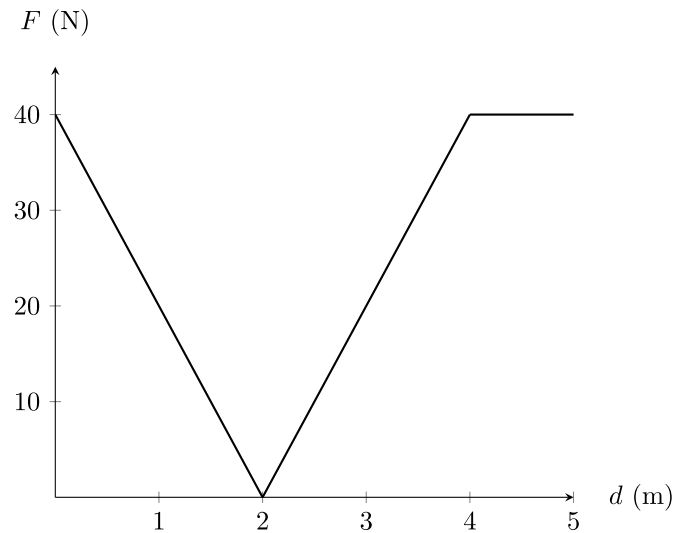
Solution

- a. $W = mad = (5 \text{ kg}) (9.8 \text{ m s}^{-2}) (4 \text{ m}) = 196 \text{ J}$.
- b. $W = Fd = (5 \text{ lb}) (4 \text{ ft}) = 20 \text{ ft-lb}$.

6.4.2. Non-Constant Force

Example 6.4.2

The graph of a force function (in Newtons) is shown below. How much work is done by the force in moving an object a distance of 5 m?



Solution

$$\begin{aligned} W &= Fd \\ &= \text{area under curve} \\ &= A_1 + A_2 + A_3 \end{aligned}$$

Then $A_1 = 2(40)\left(\frac{1}{2}\right) = 40$ J, $A_2 = 2(40)\left(\frac{1}{2}\right) = 40$ J, and $A_3 = 1(40) = 40$ J.

Altogether, the work is

$$\begin{aligned} W &= A_1 + A_2 + A_3 \\ &= 120 \text{ J} \\ &= \int_0^5 F(x) \, dx \end{aligned}$$

Example 6.4.3

A uniform cable 30 ft long and weighing 60 lb hangs vertically from a pulley system at the top of a building. A steel beam weighs 500 lb and is attached to the end of the cable. Set up and evaluate an integral to find the work required to pull the beam to the top. Explicitly write out the integral as a limit of Riemann sums.

Solution

We will consider the cable and the steel beam separately. First, the work to get the steel beam to the top of the building is $W_B = Fd = 500(30) = 15\,000$ ft-lb.

As we pull the cable up, less of it is pulling back, so the force is changing as we lift more and more cable. To account for this, consider chopping the cable up into n tiny segments of length $\Delta x = \frac{30}{n}$ ft. The i th segment is x_i ft from the top of the building. The weight (force) of each segment is the cable density times the length of the segment:

$$\begin{aligned} F_i &= \frac{60 \text{ lb}}{30 \text{ ft}} \Delta x \\ &= 2\Delta x \text{ lbs} \end{aligned}$$

The work done to lift the i th segment of cable to the top of the building is then $W_i = F_i x_i = 2\Delta x x_i$.

Next, we add up the work for all n segments to get an approximation for the total work done to lift the cable to the top of the building:

$$\begin{aligned} W_C &\approx \sum_{i=1}^n W_i \\ &\approx \sum_{i=1}^n 2\Delta x x_i \end{aligned}$$

To get the exact amount of work, we take the limit as the number of cable segments goes to infinity,

$$\begin{aligned} W_c &= \lim_{n \rightarrow \infty} \sum_{i=1}^n 2\Delta x x_i \\ &= \int_0^{30} 2x \, dx \end{aligned}$$

The work for the cable is then $W_C = \int_0^{30} 2x \, dx = x^2|_0^{30} = 900 \text{ ft-lb}$.

Combined with the beam, the total work done to lift the beam and cable to the top of the building is $900 + 15000 = 15900 \text{ ft-lb}$.

Exercise 6.4.1

Assume the previous example's set up. Find the work done to lift the beam only 10 ft up.

Theorem 6.4.1 Hooke's Law

The force required to compress or stretch a spring by x units short of or beyond its natural length is proportional to x

$$F = kx$$

Note that k is called the spring constant.

Exercise 6.4.2

Using what you know about the units for force and distance, derive the units for the spring constant in both metric and imperial units.

Example 6.4.4

A spring has a natural length of 50 cm. A force of 50 N stretches the spring to a length of 60 cm.

1. What force is needed to stretch the spring x meters?
2. How long is the spring when stretched by a force of 200 N?
3. Set up an integral to find the work done in stretching the spring 20 cm.
4. Set up an integral to find the work done in stretching the spring from a length of 100 cm to 110 cm.
5. 20 J of work is done in stretching the spring a certain distance. How far was the spring stretched?

Solution

1. In general, $F = kx$, so if we know k we can get a formula for the force needed to stretch the spring x meters. Going from 50 cm to 60 cm is a change of 0.1 m, since this takes 50 N we know

$$\begin{aligned}50 \text{ N} &= k(0.1 \text{ m}) \\k &= \frac{50 \text{ N}}{0.1 \text{ m}} \\k &= 500 \text{ N m}^{-1}\end{aligned}$$

Then a formula for force needed to stretch this spring x meters is $F = 500x$.

2. Using our formula and $F = 200$,

$$\begin{aligned}F &= 500x \\200 &= 500x \\x &= 0.4 \text{ m}\end{aligned}$$

Remember that this is how far from the spring rest length it is, so we have to add the 0.5 m to this to get that the spring is 0.9 m.

3. 20 cm is 0.2 m. Then the work is

$$\begin{aligned}W &= \int_a^b F(x) \, dx \\&= \int_0^{0.2} 500x \, dx\end{aligned}$$

4. The trick here is finding the bounds of the integral. 100 cm is 1 m and 110 cm is 1.1 m. We also need to know how far from rest length these distances are: $1 - 0.5 = 0.5$ m and $1.1 - 0.5 = 0.6$ m. Then, the work done is

$$W = \int_{0.5}^{0.6} 500x \, dx$$

5. Let l be how far the spring was stretched (from rest). Then the work to get it there is

$$W = \int_0^l 500x \, dx = 20 \text{ J}$$

Then solving for l

$$20 = \int_0^l 500x \, dx$$

$$20 = 250x^2 \Big|_0^l$$

$$20 = 250l^2$$

$$l^2 = \frac{2}{25}$$

$$l = \pm \frac{\sqrt{2}}{5}$$

The spring was stretched an additional $\frac{\sqrt{2}}{5}$ m.

6.4 a Section Summary:

- We used integrals and Riemann sums to describe work done to lift heavy objects and stretch springs.